

Supplemental Material for “Magnetic bimerons as skyrmion analogues in in-plane magnets”

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In this Supplemental Material we prove or numerically validate several claims made in the main text, which were motivated by geometric considerations. In addition, we present results of atomistic and micromagnetic simulations for different parameters and setups.

I. BIMERON CRYSTALS IN FRUSTRATED MAGNETS

The results of atomistic Monte Carlo simulations presented here validate the claim that in triangular lattices with frustrated exchange interactions bimeron crystals can be stabilized. We use nearest-neighbor exchange $J_1 = 0.438 J$, second-nearest-neighbor exchange $J_2 = -0.136 J$, and third-nearest-neighbor exchange $J_3 = J_2$; here J is an arbitrary unit set to 1 in the simulations. For both magnetic field $\mathbf{B}$ and easy-axis anisotropy along z we find skyrmion crystals, whereas for both quantities along x bimeron crystals are stabilized (see Fig. S1).

As in the phase diagram for skyrmion crystals in Ref. 1, the topologically nontrivial phase [Fig. S1(b)] is stabilized for field strengths slightly below those stabilizing the ferromagnetic phase [see Fig. S1(d)]. At the transition between the two phases a mixed state shows up: an in-plane ferromagnet with few isolated bimerons is stabilized [see Fig. S1(c)]. The bimeron crystal phase does not extend down to $B = 0$: there a spin-spiral phase is observed [see Fig. S1(a)].

In the main text we focused on bimerons as analogues of skyrmions after a global spin rotation with easy-axis anisotropy. In Fig. S2 we show that easy-plane magnets can host bimerons as well. Recall that in systems with an easy-plane anisotropy, all in-plane directions are equivalent. The presence of the magnetic field, however, breaks the C_{6v} symmetry, with the consequence that bimerons can arise.

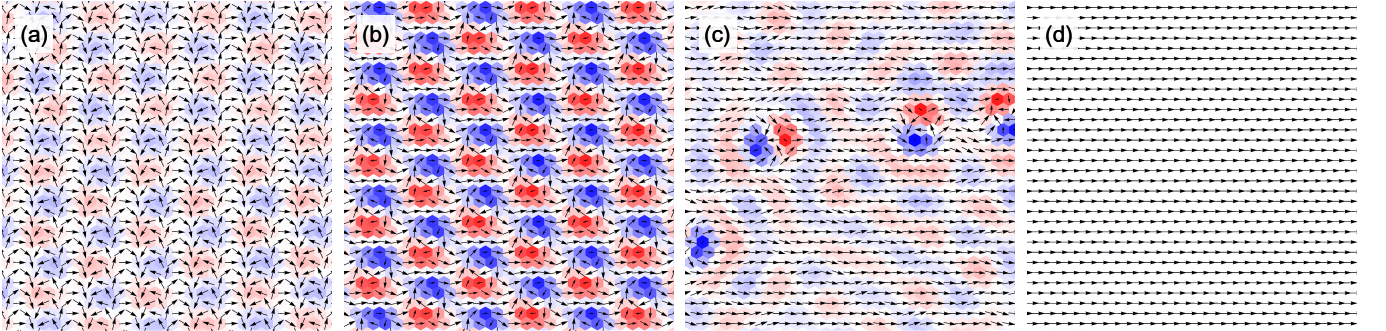


FIG. S1. Textures in frustrated magnets with easy-axis anisotropy: (a) spiral phase, (b) bimeron crystal phase, (c) ferromagnet with few individual bimerons, and (d) ferromagnet. The exchange parameters are given in the text. Magnetic fields and anisotropies are (a) $B_x = K_x = 0.1 J$, (b) $B_x = K_x = 0.25 J$, (c) $B_x = K_x = 0.32 J$, and (d) $B_x = K_x = 0.4 J$. The cluster consists of 32×32 magnetic moments.

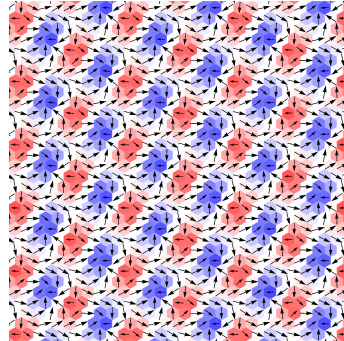


FIG. S2. Bimerons in frustrated magnets with easy-plane anisotropy. Parameters as in Fig. S1 except for $\mathbf{B} = 0.25 J \mathbf{e}_x$ and $\mathbf{K} = -0.2 J \mathbf{e}_z$.

II. TOPOLOGICAL HALL EFFECT IN BIMERON CRYSTALS

We briefly summarize the discussion of the topological Hall effect (THE) of electrons in a skyrmion crystal, as presented in Ref. 2, and carry the discussion over to bimeron crystals. We consider the tight-binding Hamiltonian

$$H = t \sum_{\langle i,j \rangle} c_i^\dagger c_j + m \sum_i \mathbf{s}_i \cdot (c_i^\dagger \boldsymbol{\sigma} c_i), \quad (1)$$

which describes the electrons by means of creation $c_i^\dagger$ and annihilation c_i operators on the site i that interact via their spin with a magnetic texture $\{\mathbf{s}_i\}$. The first term models the hopping of the conduction electrons. For simplicity only nearest-neighbor hopping with amplitude t is considered. The second term describes Hund's coupling of electron spins to the magnetization, here $\boldsymbol{\sigma}$ is the vector of Pauli matrices. This interaction is quantified by m . The texture itself is periodic (either a skyrmion or a bimeron crystal) with a unit cell consisting of n atoms. Considering only s conduction electrons, this leads to $2n$ bands.

For $m = 0$ the band structure is that of the system without skyrmions or bimerons but back-folded because of the larger magnetic unit cell. Each band is spin degenerate. Prominent features of this initial band structure are determined by the structural lattice. A triangular structural lattice gives the band structure depicted in Fig. 4(a) and (b) of Ref. 2. This band structure has minima at $E = -3t$, a maximum at $E = +6t$, and a van Hove singularity (VHS), i. e., a saddle point in the band structure, at $E = -2t$. Between the minima and the VHS a constant energy cut (Fermi line at the particular energy) yields two electron pockets; between the VHS and the maximum there is one hole pocket.

For positive coupling $m > 0$ the spin degeneracy is lifted and the electron spins align (partially) with the texture. For considerably large coupling (e. g. $m > 5t$) the system is in the adiabatic limit, in which the electron spin is almost completely aligned with the texture. This alignment is accompanied by a block separation: the two blocks in the band structure – one for parallel and the other for antiparallel spin alignment – look similar (apart from minor details) and are shifted by $\pm 5t$. Due to the alignment of the electron spin with the topologically nontrivial spin texture, the bands become topologically nontrivial themselves, with the consequence of an emergent Hall conductivity³

$$\sigma_{xy}(E_F) = -\frac{e^2}{h} \frac{1}{2\pi} \sum_n \int_{\text{BZ}} \Omega_n^{(z)}(\mathbf{k}) f(E_n(\mathbf{k}) - E_F) d^2k, \quad (2)$$

calculated from the Berry curvature

$$\Omega_n^{(z)}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{\langle u_n(\mathbf{k}) | \partial_{k_x} H(\mathbf{k}) | u_m(\mathbf{k}) \rangle \langle u_m(\mathbf{k}) | \partial_{k_y} H(\mathbf{k}) | u_n(\mathbf{k}) \rangle}{[E_n(\mathbf{k}) - E_m(\mathbf{k})]^2}, \quad (3)$$

where f is the Fermi distribution function.

The transverse conductivity for the two blocks looks similar, but with the opposite sign (due to the opposite spin alignment, see Fig. S3). Within a block the conductivity has opposite signs for energies below and above the VHS. This change of sign is attributed to the electron- or hole-like character of the Fermi lines. The conductivity is quantized in units of e^2/h within the global band gaps, which are small for the chosen skyrmion size of $n = 108$ lattice sites per skyrmion. The curve appears therefore rather smooth. More details on Chern numbers and the emergent field are given in Ref. 2.

Exchanging the skyrmion texture with that of a bimeron crystal of the same size and the same periodicity, i. e., all the spins are merely rotated by 90° around the y axis, the conductivity remains unchanged (Fig. S3). This is explained by the fact that the spin texture in a *local* (site-dependent) coordinate system (e. g. spin along the local z axis) remains invariant under global spin rotations. Therefore, the accumulated Berry phase remains invariant as well. In Refs. 2 and 4 it has been shown that the THE can effectively be described as a quantum Hall effect using the emergent field of a skyrmion as orbital field. This field is a fictitious quantity which points out-of-plane and is proportional to the topological charge density. Since the latter is invariant under global spin rotations the results of Fig. S3 are well understood.

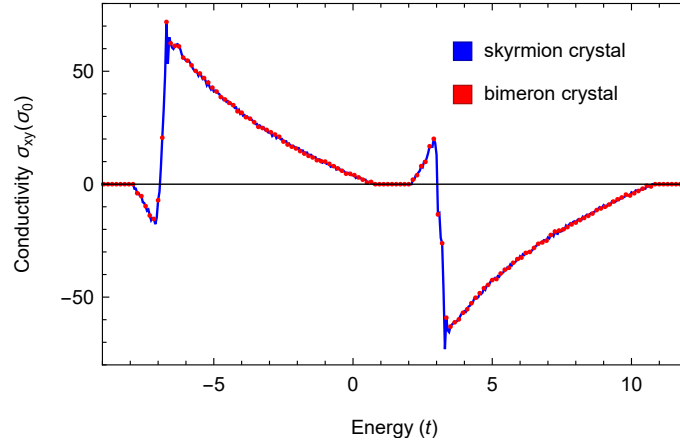


FIG. S3. Topological Hall conductivity vs. location of the Fermi energy of a bimeron crystals (red) and a skyrmion crystal (blue). The spectra, both calculated for $n = 108$ sites and $m = 5t$, are identical.

III. LANDAU-LIFSHITZ-GILBERT EQUATION

For the atomistic simulations of stabilizing isolated skyrmions, bimerons, and antiskyrmions in ferromagnets [Fig. 1(e-g) of the main text], as well as for the micromagnetic simulations (Fig. 2 of the main text) we solved the Landau-Lifshitz-Gilbert equation for each discretized micromagnetic moment $\mathbf{m}_i$ ⁵⁻⁷

$$\dot{\mathbf{m}}_i = -\gamma_e \mathbf{m}_i \times \mathbf{B}_{i,\text{eff}} + \alpha \mathbf{m}_i \times \dot{\mathbf{m}}_i + \gamma_e \frac{j\hbar P}{2ed_z M_s} [(\mathbf{m}_i \times \mathbf{s}_{\text{in}}) \times \mathbf{m}_i]. \quad (4)$$

The first term on the right-hand side describes the precession of the magnetic moment $\mathbf{m}_i$ around the effective magnetic field $\mathbf{B}_{i,\text{eff}}$, which arises due to the interactions with neighboring magnetic moments and the external magnetic field; $\gamma_e = \gamma/\mu_0 = 1.760 \times 10^{11} \text{ T}^{-1}\text{s}^{-1}$ is the gyromagnetic ratio of an electron. The second term describes dissipation, where α is the Gilbert damping constant.

The third term, the so-called ‘in-plane spin torque’, describes interaction of the magnetic moments with injected spins $\mathbf{s}_{\text{in}}$; j is the applied current, d_z is the thickness of the ferromagnetic layer and M_s the saturation magnetization. The prefactors are deliberately written such to bring the SOT scenario to mind, in which P is the spin-polarization of the perpendicularly injected spin current in scenario (i) or is given by the spin Hall angle (charge to spin conversion) in scenario (ii). μ_0 is the vacuum permeability.

The ‘out-of-plane’ or ‘field-like’ torque $\propto \mathbf{m}_i \times \mathbf{s}_{\text{in}}$ has been dropped, since it does not drive but slightly distorts the quasiparticles. Recently two other torque terms have been identified⁸ whose effects on bimerons and also skyrmions have not sufficiently been investigated yet. Their discussion is beyond the scope of this paper. The parameters chosen for the simulations are presented in Table I.

Quantity	Value	Unit
Saturation magnetization M_s	0.58	MA/m
Exchange stiffness J	15	pJ/m
DMI strength D	3	mJ/m ²
Uniaxial anisotropy $ K $	0.8	MJ/m ³
Gilbert damping α	0.3	
Polarization or Spin Hall angle P	0.4	
Racetrack thickness d_z	0.5	nm
Racetrack width	40	nm

TABLE I. The parameters used in all simulations. The track is periodic in x direction. A length of 500 nm is simulated.

IV. THIELE EQUATION ANALYSIS

The Thiele equation⁹⁻¹³ for the SOT scenario

$$b\mathbf{G} \times \mathbf{v} - bD\alpha\mathbf{v} + Bj\mathbf{I}\mathbf{s}_{\text{in}} = 0. \quad (5)$$

allows for a center-of-mass description of magnetic quasiparticles (domain walls, skyrmions, and also bimerons). Here $\mathbf{v}$ is the particle's velocity, $b = M_s d_z / \gamma_e$, and $B = \hbar P / (2e)$. Note that $\mathbf{s}_{\text{in}}$ (as $\mathbf{m}_i$) is a unit vector. This equation is derived by integrating the LLG equation over the spacial extension of the quasiparticle, thereby condensing all properties in the gyromagnetic coupling vector

$$\mathbf{G} = -4\pi N_{\text{Sk}} \mathbf{e}_z, \quad (6)$$

the dissipation tensor with elements

$$D_{ij} = \int \partial_{x_i} \mathbf{m}(\mathbf{r}) \cdot \partial_{x_j} \mathbf{m}(\mathbf{r}) d^2 r \quad (7)$$

(only $D_{xx} = D_{yy}$ are nonzero for a Néel skyrmion), and the tensor $\mathbf{I}$ with elements

$$I_{ij} = \int [\partial_{x_i} \mathbf{m}(\mathbf{r}) \times \mathbf{m}(\mathbf{r})]_{x_j} d^2 r. \quad (8)$$

The latter accounts for the action of the injected spins on the texture. Only $I_{xy} = -I_{yx}$ are nonzero for a Néel skyrmion.

Under a global spin rotation (from skyrmion to bimeron) $\mathbf{G}$ and $\mathbf{D}$ remain invariant. The tensor $\mathbf{I}$ now only has $I_{xy} = I_{yz}$ different from zero. For the skyrmion and bimeron from Fig. 1(a) from the main text $N_{\text{Sk}} = -1$, $D_{xx} > 0$, and $I_{xy} > 0$. Comparing the components of the Thiele equation for the skyrmion,

$$4\pi b \begin{pmatrix} -v_y \\ v_x \\ 0 \end{pmatrix} = bD_{xx}\alpha \begin{pmatrix} v_x \\ v_y \\ 0 \end{pmatrix} - BjI_{xy} \begin{pmatrix} s_y \\ -s_x \\ 0 \end{pmatrix}, \quad (9)$$

and for the bimeron,

$$4\pi b \begin{pmatrix} -v_y \\ v_x \\ 0 \end{pmatrix} = bD_{xx}\alpha \begin{pmatrix} v_x \\ v_y \\ 0 \end{pmatrix} - BjI_{xy} \begin{pmatrix} s_y \\ s_z \\ 0 \end{pmatrix}, \quad (10)$$

shows that skyrmions and bimerons are driven identically if the injected spins are rotated in the same way as the textures; here the rotation is around $\mathbf{y}$. This means in particular that skyrmions are propelled for injected spins $\mathbf{s}_{\text{in}} \perp \mathbf{z}$, whereas bimerons are driven by those with $\mathbf{s}_{\text{in}} \perp \mathbf{x}$. For $\mathbf{s}_{\text{in}} \parallel \mathbf{y}$, the translation of skyrmions and bimerons is identical: The second line of the two above equations tells the skyrmion Hall angle $\theta_{\text{SK}} = \arctan(v_y/v_x) = \arctan[4\pi/(D_{xx}\alpha)]$ for this case. This finding is in good agreement with the results from the main text (cf. orange lines in Fig. 3).

V. BIMERONS IN EASY-PLANE RACETRACKS

Figure S4 shows that bimerons can be stabilized in racetracks with an easy-plane anisotropy, as long as a magnetic field is applied along x . The bimeron moves along the track for $\mathbf{s}_{\text{in}} \parallel \mathbf{y}$, similar to the bimerons discussed in the main text. Also the velocities are almost identical.

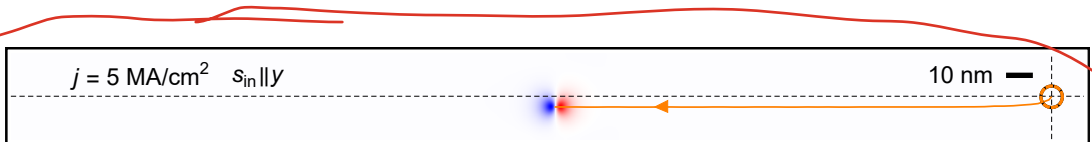


FIG. S4. Current-driven motion of a bimeron in an easy-plane magnet. As Fig. 3(c) of the main text but for different parameters ($K_z = -0.8 \text{ MJ/m}^3$, $B_x = 1 \text{ T}$).

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